

9-2e Comparing Depreciation Methods

The three depreciation methods are summarized in [Exhibit 8](#). All three methods allocate a portion of the total cost of an asset to an accounting period, while never depreciating an asset below its residual value.

Exhibit 8

Summary of Depreciation Methods

Method	Useful Life	Depreciable Cost	Depreciation Rate	Depreciation Expense
Straight-line	Years	Cost less residual value	Straight-line rate	Constant
Units-of-activity	Units of activity	Cost less residual value	$\frac{\text{Cost} - \text{Residual value}}{\text{Total units of activity}}$	Variable
Double-declining-balance	Years	Declining book value, but not below residual value	Straight-line rate $\times 2$	Declining

The straight-line method provides for the same periodic amounts of depreciation expense over the life of the asset. The units-of-activity method provides for periodic amounts of depreciation expense that vary, depending on the amount the asset is used. The double-declining-balance method provides for a higher depreciation amount in the first year of the asset’s use, followed by declining amounts.

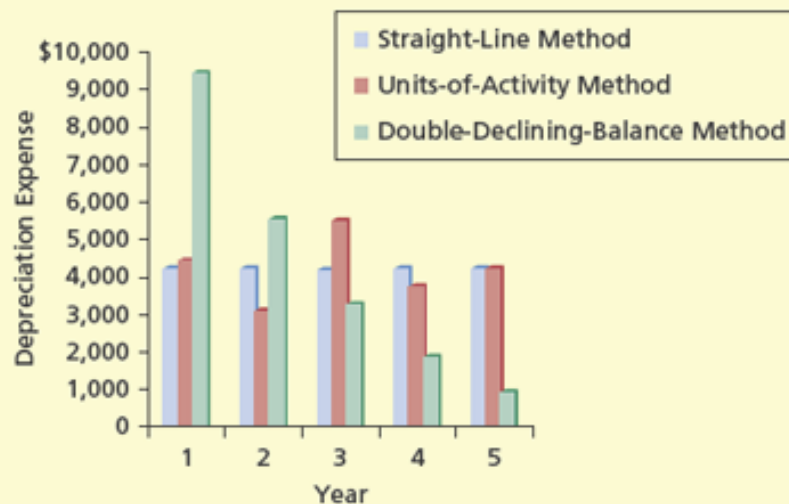
[Exhibit 9](#) illustrates depreciation expense for each depreciation method over the five-year life of the forklift.

Exhibit 9 Comparing Depreciation Methods

Year	Depreciation Expense		
	Straight-Line Method	Units-of-Activity Method	Double-Declining-Balance Method
1	\$ 4,400*	\$ 4,620 (\$2.20 × 2,100 hrs.)	\$ 9,600.00 (\$24,000 × 40%)
2	4,400	3,300 (\$2.20 × 1,500 hrs.)	5,760.00 (\$14,400 × 40%)
3	4,400	5,720 (\$2.20 × 2,600 hrs.)	3,456.00 (\$8,640 × 40%)
4	4,400	3,960 (\$2.20 × 1,800 hrs.)	2,073.60 (\$5,184 × 40%)
5	4,400	4,400 (\$2.20 × 2,000 hrs.)	1,110.40**
Total	<u>\$22,000</u>	<u>\$22,000</u>	<u>\$22,000.00</u>

*\$4,400 = (\$24,000 – \$2,000) ÷ 5 years

**\$3,110.40 – \$2,000.00 because the equipment cannot be depreciated below its residual value of \$2,000.



Check up Corner 9-1

Fixed Assets: Depreciation

On the first day of the year, Firefall Company acquired equipment for use in operations at a cost of \$340,000. The equipment was expected to have a useful life of 4 years or 1,000 hours, and a residual value of \$20,000. The equipment was used for 280 hours during the first year, 260 hours during the second year, 240 hours during the third year, and 220 hours during the fourth year.

Determine the annual depreciation expense in each year and the book value of the equipment at the end of each year under the:

1. straight-line method
2. units-of-activity method
3. double-declining-balance method

Chapter 9: Long-Term Assets: Fixed and Intangible: 9-2e Comparing Depreciation Methods

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